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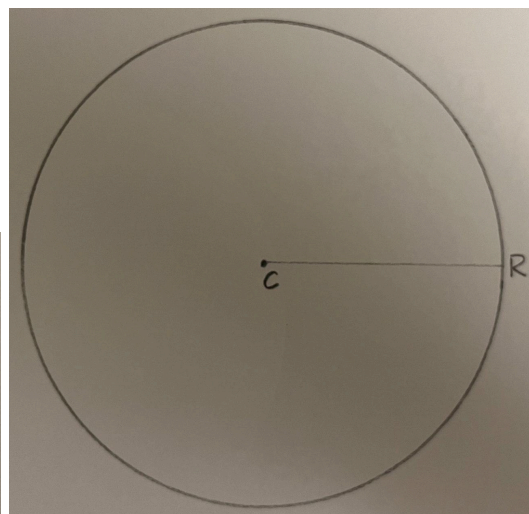
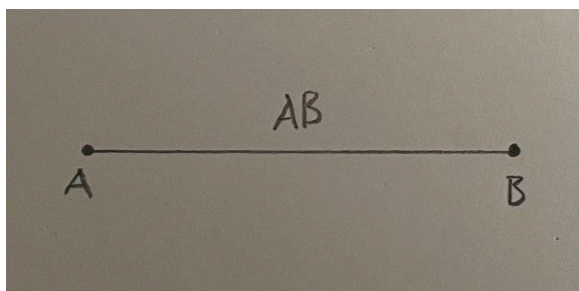
Abstract Algebra 1

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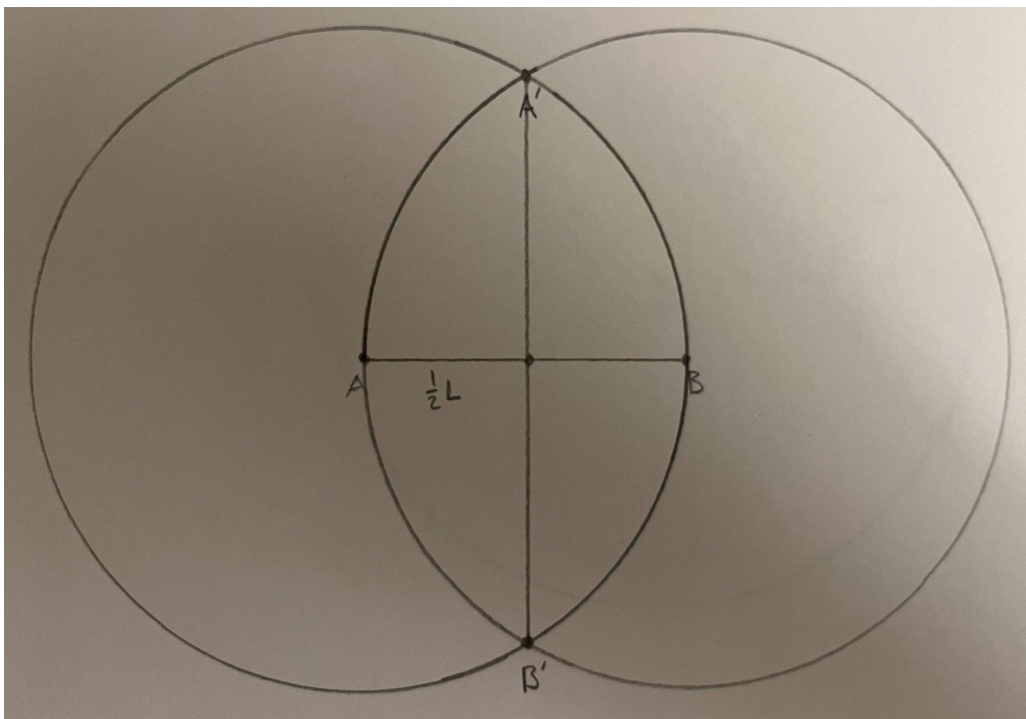
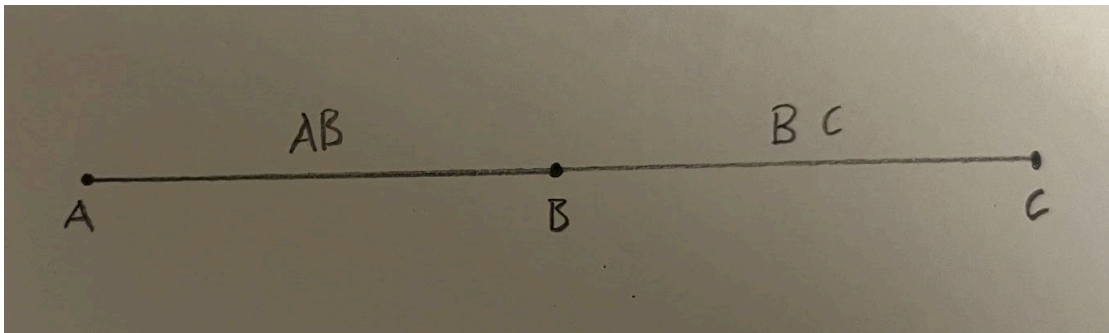
Geometric Construction of The Flate Irons Boulder

A brief exploration into the subject of geometric construction, some times known as euclidean construction. Geometric construction was quite popular among Ancient Greek mathematicians. The art of geometric construction comes from constructing a geometric figure obeying the following limitation. The allowed tools are a compass and an unmarked straight edge in addition to an independently defined unit length. It is from this unit length that the construction must be attainable using the aforementioned tools, straight edge, and compass.

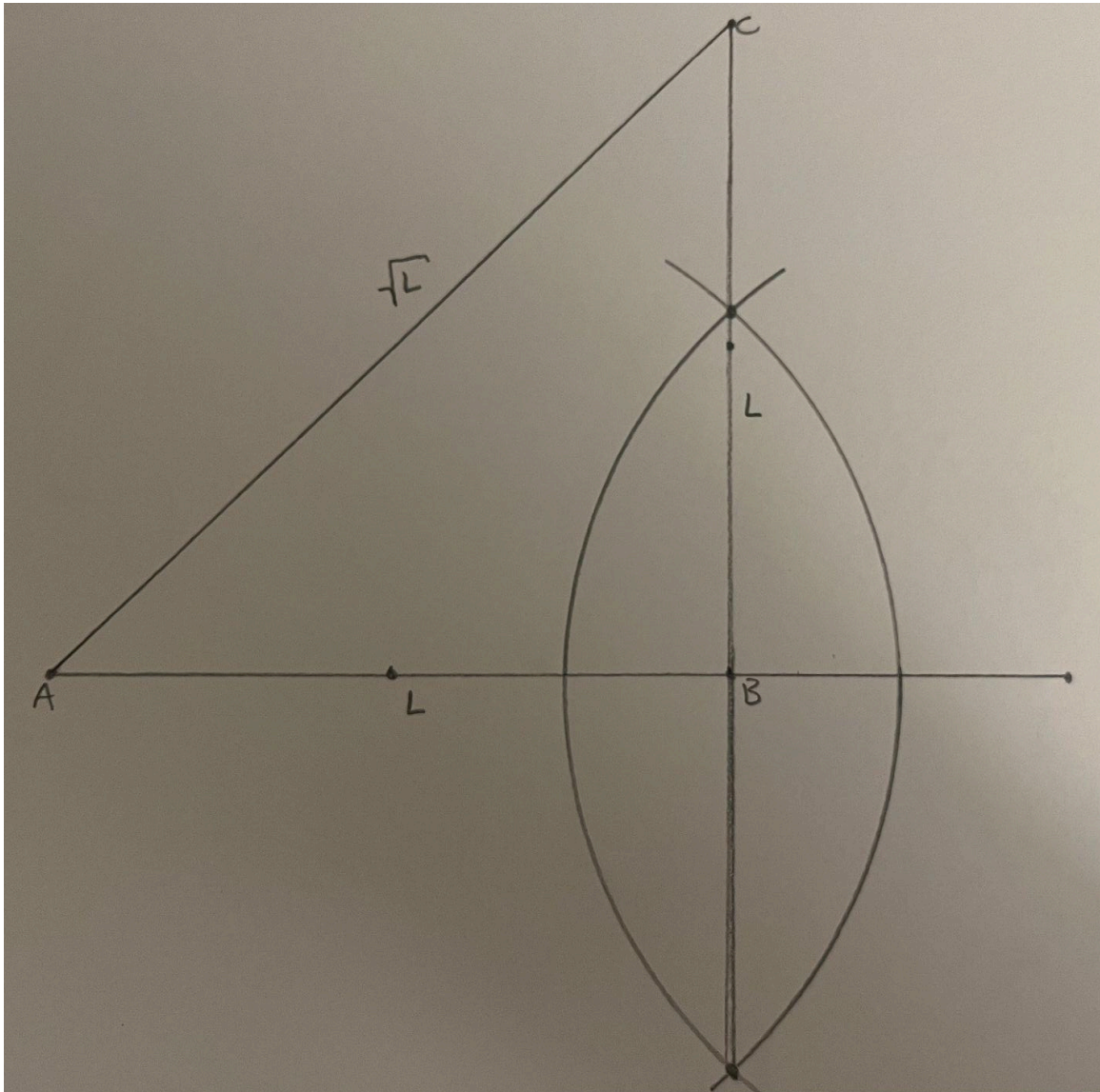
There are certain shapes easily created such as a circle or line. Both are defined by two points. For the straight line if A and B are points the line AB is the shortest distance along a surface between A and B. For the purposes of this subject, we restrict ourselves to Euclidean surfaces (i.e. no spheres, hyperboloids, or other non-flat surfaces). To construct the defined line plot points A and B place the straight edge such that it crosses both points A and B, and then trace along the straight edge. The second basic construction is a circle, also defined by two points C and R. C is the center of the circle on the page. While, R isn't really a point and more a collection of points. R is the radius of the circle. This is the collection of points of length R from the center C. To construct this simply place the fixed point of the compass on C then measure R in terms of the unit length defined and trace the circle by rotating the drawing end of the compass about the fixed end.



This brings up an important question. How do we construct lengths other than the defined unit length? There are many approaches but this paper will only offer a few tricks. Let L be the unit length in question. Then $2L$ is obtained simply by placing points A , B , C in equal distances such that the distance between A and B is L and the distance between B and C is L then draw the line AC such that AC passes through A , B , and C to construct a length of $2L$. This process can be repeated n times where n is an integer to achieve any length of the form nL . Next to construct a line of length $\frac{1}{2}L$ start by constructing a line of length L , then draw two circles of radius R such that R is the distance of length L . If the line of length L is AB the center of the first circle is A and the center of the second circle is B . Next draw a line $A'B'$ such that A' and B' are the intersection points of the two circles. This line $A'B'$ will intersect AB at precisely the midpoint. Then the line of length $\frac{1}{2}L$ is the distance between the points A and the intersection between AB and $A'B'$. There is a second unmentioned result of this process. The line $A'B'$ is perpendicular to AB (i.e. AB and $A'B'$ intersect and a 90° angle).



There are many more techniques to explore but this paper will only mention one more. To construct a slope at an angle θ we begin by constructing a line AB of length n . Then use the above techniques to construct a line BC of length m such that AB is perpendicular to BC. Next, complete a right triangle ABC by drawing the line AC. Notice the choice of θ is limited by the constructible lengths AB and BC.



Building on the idea of construction, in the study of field theory a number x is constructible if a length of x is constructible using geometric construction. From plain geometry as soon as x and y exist as constructible numbers a few more constructible numbers exist. If x and y are constructible numbers then so too is $x+y$, $x-y$, $y-x$, $x*y$, and assuming y is not $0 \frac{x}{y}$. This satisfies the criteria for constructible numbers to be a field. A field is similar to a group in the sense both a field and group are defined off sets. But where a group is coupled with one binary operation that satisfies the following:

- The operation is associative.
- There is an element e in the set such that performing the operation on an element a in the set either the left or right archives $ae = ea = a$.
- For every element a in the set there is an element a^{-1} in the set such that $aa^{-1} = a^{-1}a = e$.

Where a group has one binary operation a field has two binary operations that satisfy the three preceding criteria. It is important to note that while an additive identity and inverse are required for a ring, unlike a group there need not be a multiplicative identity or multiplicative inverse. Thus the constructable numbers are a field under addition and multiplication, and have a multiplicative identity and inverse.

We can use this to confirm what the ancient greek mathematicians long thought to be unsolvable problems (geometric constructions) in the study of geometric construction. Before introducing the problems it is worth noting these problems came about due to the greeks in there studies of geometric construction, and after large unsuccessful attempts to construct certain things the greeks believed these problems were unsolvable but lacked the necessary tools to prove this rigorously. The proof for the unsolvability of these problems came in with the exploration into the study of field theory.

The first problem was to double the volume of a cube. To do this with geometric construction a length of $\sqrt[3]{L}$ must be constructed. However, from Fermat's Last Theorem we know the equation $x^n + y^n = z^n$ has no integer solutions x, y, z where n is greater than or equal to 3. Thus our triangular method from before or any other method will fail. This is not conclusive proof as the field of constructable numbers expands far beyond the integers, as $\sqrt{L}$ is constructed using the triangle method above where AB is the same length of BC , L . But the argument does offer inclination as to $\sqrt[3]{L}$ is not constructable as Fermat's Last Theorem suggests geometrically there is no construction of $\sqrt[3]{L}$ where as there is a construction of $\sqrt{L}$.

The second problem is quit troubling as it was posited by ancient greek mathematicians that it is impossible to trisect an angle using geometric construction. This is troubling as developing the angles of the Flat Irons required a lot of angle varying techniques. But frist again field theory offers satisfaction to greek mathematicians in their suspensions of trisecting an angle to be impossible.

Proof

If it were possible to trisect an angle of 60° we could construct a point $(A,0)$ where $A = \cos(20^\circ) \Rightarrow$ we count construct $(B,0)$ where $B = 2 \cos(20^\circ)$ which generalizes the construction of trisecting an angle. However, from elementary trigonometry

$$\begin{aligned}\cos(3\theta) &= 4\cos^3(\theta) - 3\cos(\theta) \\ \Rightarrow \cos(60^\circ) &= 4\cos^3(20^\circ) - 3\cos(20^\circ)\end{aligned}$$

Thus $B = 2 \cos(20^\circ)$ satisfies the polynomial $B^3 - 3B - 1 = 0$. But the polynomial

$$p(x) = x^3 - 3x - 1$$

is irreducible over the rational numbers by Einsteins criteria thus this exists in a field that contains powers of 3 but the constructable points exist in a field of at most power 2 $\Rightarrow$ contradiction.

QED

This proof was presented in Charles A. Pinter's *A Book of Abstract Algebra Second Edition*.

Now returning to constructing the Flat Irons. This process was done in several parts. First an omission, a marked ruler was used with the unit length being 1 cm. However, in staying true to geometric construction once the 1 cm unit length was drawn the marking on the ruler were ignored. The Flatirons were constructed one at a time being independent of one another with the rolling hill leading up to them as the connection. This connection point is what lead to the difficulty with angles as none of the Flat Irons are at a 90° in relation to the connection point. This was solved using several varying angle producing techniques including the triangle method previously stated rather than using division of angles. Starting with the construction of the rolling hills, connected circular segments and a few lines were produced. Then came the challenge of constructing the Flat Irons. Each one was constructed from a single sloped line in relation to the previously constructed rolling hills. Then once the location of each Flat Iron was constructed along with its identifying line the rest of the Flat Iron was built around this identifying line using various techniques. But in each case all straight lines were constructed first then each circular arc portion of the flatirons was filled in upon the completing of the edge work. Finally, the skyline trees and detail on the rolling hills was filled in again using a variety of techniques.

The result is the image presented with this paper.

